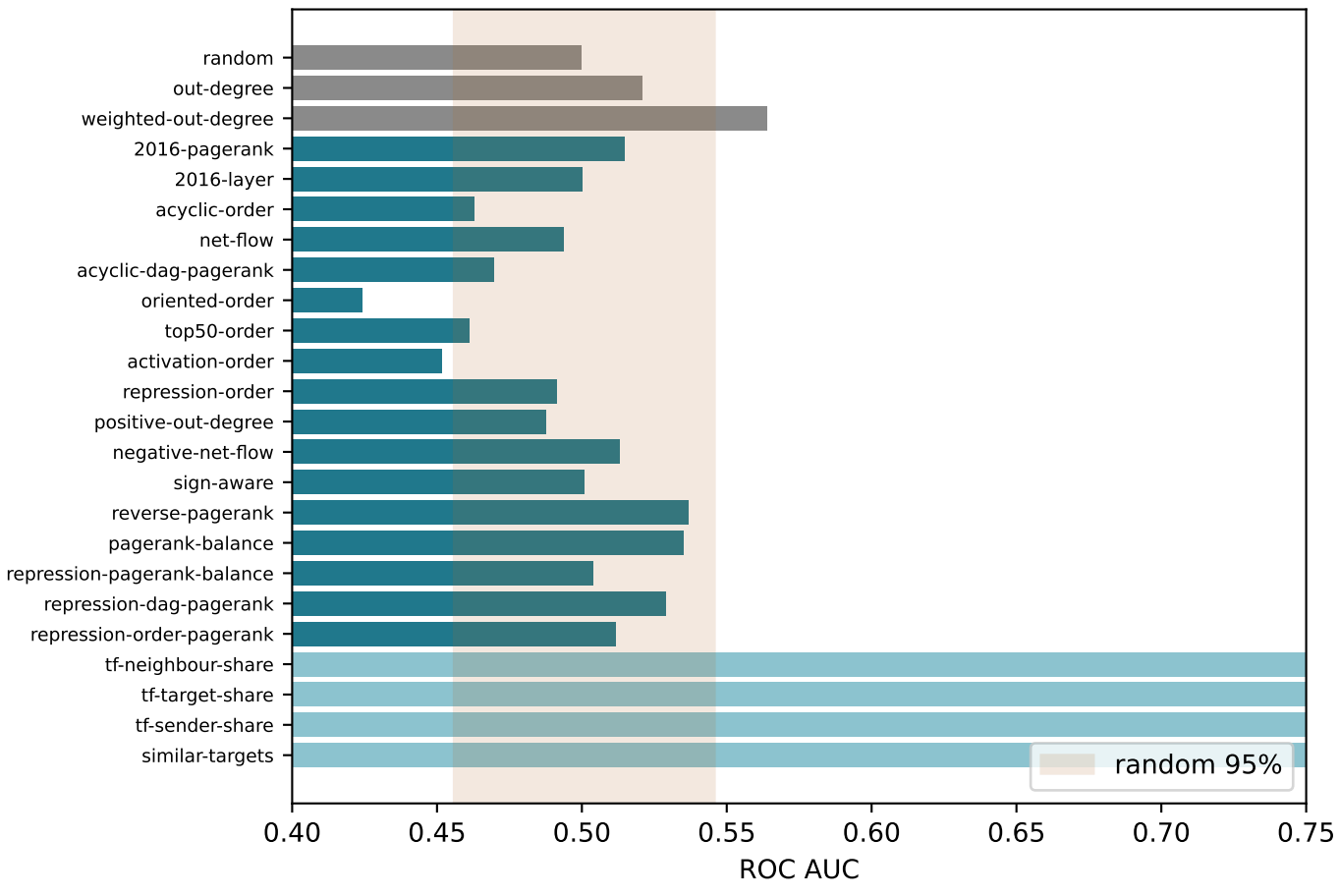
Regulator ranking against reference 'dream5-tfs'

2026-11-08 · 1,643 genes ranked, 195 in tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	11.8 (6–18)	0.500 (0.456–0.546)				
out-degree	8	0.521	0.920	0.180	-0.043 (-0.055, -0.032)	< 0.001
weighted-out-degree	10	0.564	0.761	0.002	+0.000 (+0.000, +0.000)	
2016-pagerank	11	0.515	0.642	0.254	-0.049 (-0.077, -0.019)	< 0.001
2016-layer	10	0.500	0.761	0.485	-0.064 (-0.095, -0.030)	< 0.001
acyclic-order	7	0.463	0.959	0.954	-0.101 (-0.125, -0.077)	< 0.001
net-flow	7	0.494	0.959	0.605	-0.070 (-0.090, -0.049)	< 0.001
acyclic-dag-pagerank	9	0.470	0.854	0.907	-0.094 (-0.121, -0.070)	< 0.001
oriented-order	6	0.424	0.986	1.000	-0.140 (-0.170, -0.109)	< 0.001
top50-order	11	0.461	0.642	0.959	-0.103 (-0.128, -0.078)	< 0.001
activation-order	7	0.452	0.959	0.984	-0.112 (-0.141, -0.083)	< 0.001
repression-order	9	0.491	0.854	0.640	-0.073 (-0.100, -0.046)	< 0.001
positive-out-degree †	12	0.488	0.523	0.698	-0.076 (-0.102, -0.050)	< 0.001
negative-net-flow †	8	0.513	0.920	0.281	-0.051 (-0.074, -0.026)	< 0.001
sign-aware †	7	0.501	0.959	0.479	-0.063 (-0.083, -0.044)	< 0.001
reverse-pagerank †	9	0.537	0.854	0.050	-0.027 (-0.039, -0.013)	< 0.001
pagerank-balance †	9	0.535	0.854	0.059	-0.029 (-0.051, -0.003)	0.028
repression-pagerank-balance †	7	0.504	0.959	0.426	-0.060 (-0.085, -0.033)	< 0.001
repression-dag-pagerank †	11	0.529	0.642	0.096	-0.035 (-0.065, -0.007)	0.008
repression-order-pagerank †	10	0.512	0.761	0.303	-0.052 (-0.081, -0.027)	< 0.001
tf-neighbour-share ‡	93	0.856	< 0.001	< 0.001	+0.292 (+0.248, +0.338)	< 0.001
tf-target-share ‡	89	0.839	< 0.001	< 0.001	+0.275 (+0.231, +0.319)	< 0.001
tf-sender-share ‡	91	0.859	< 0.001	< 0.001	+0.295 (+0.252, +0.340)	< 0.001
similar-targets ‡	85	0.781	< 0.001	< 0.001	+0.217 (+0.169, +0.265)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).

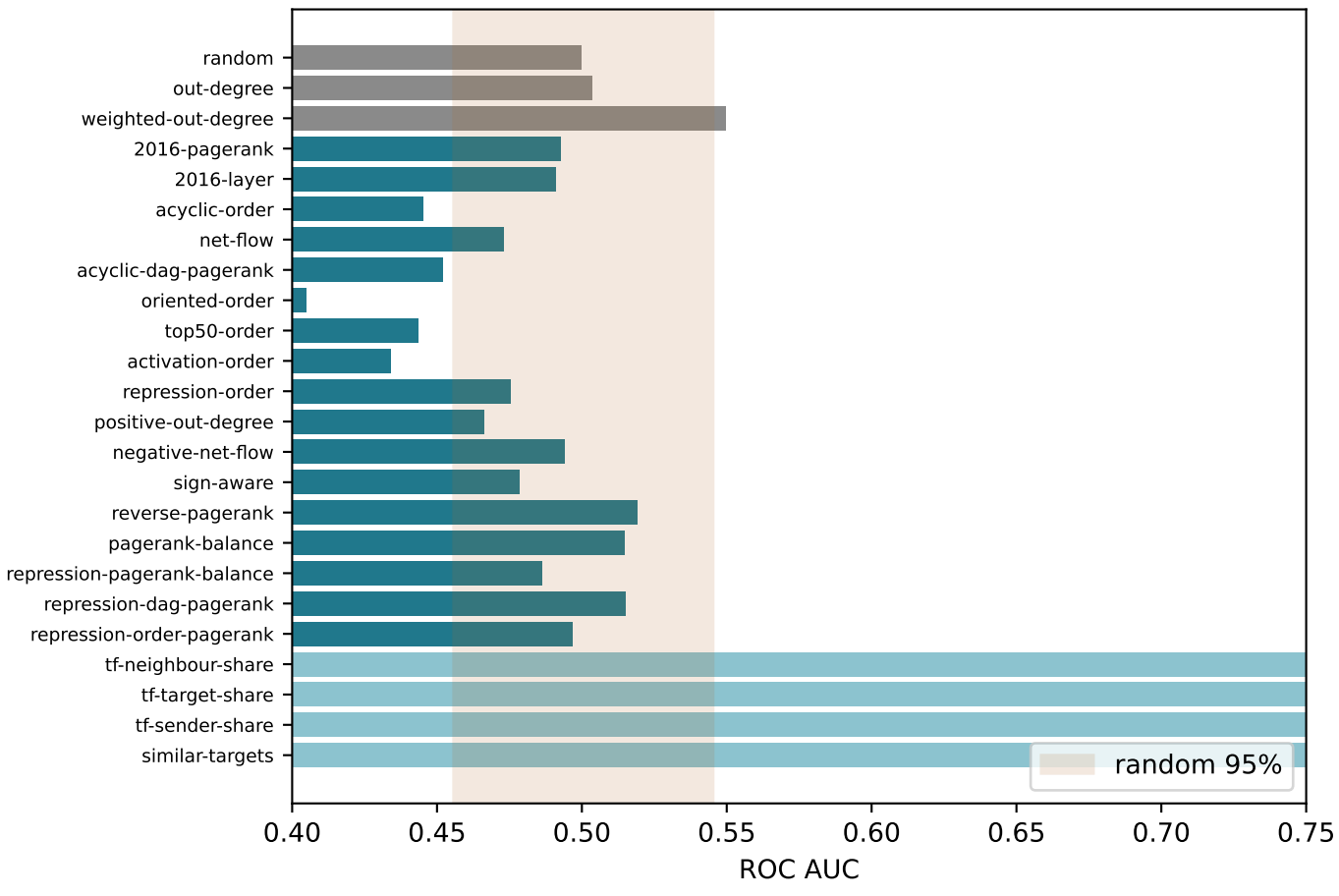


Regulator ranking against reference 'true-regulators'

2026-11-08 · 1,643 genes ranked, 178 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	10.8 (5–17)	0.500 (0.456–0.545)				
out-degree	8	0.504	0.862	0.432	-0.046 (-0.057, -0.034)	< 0.001
weighted-out-degree	10	0.550	0.643	0.015	+0.000 (+0.000, +0.000)	
2016-pagerank	11	0.493	0.509	0.620	-0.057 (-0.086, -0.025)	< 0.001
2016-layer	9	0.491	0.769	0.647	-0.058 (-0.092, -0.024)	< 0.001
acyclic-order	7	0.445	0.928	0.992	-0.104 (-0.129, -0.079)	< 0.001
net-flow	7	0.473	0.928	0.880	-0.076 (-0.097, -0.053)	< 0.001
acyclic-dag-pagerank	9	0.452	0.769	0.983	-0.098 (-0.123, -0.071)	< 0.001
oriented-order	6	0.405	0.969	1.000	-0.145 (-0.174, -0.112)	< 0.001
top50-order	10	0.444	0.643	0.994	-0.106 (-0.131, -0.080)	< 0.001
activation-order	7	0.434	0.928	0.998	-0.115 (-0.145, -0.084)	< 0.001
repression-order	8	0.475	0.862	0.850	-0.074 (-0.103, -0.047)	< 0.001
positive-out-degree †	11	0.466	0.509	0.928	-0.083 (-0.111, -0.055)	< 0.001
negative-net-flow †	7	0.494	0.928	0.591	-0.056 (-0.081, -0.029)	< 0.001
sign-aware †	7	0.479	0.928	0.820	-0.071 (-0.091, -0.050)	< 0.001
reverse-pagerank †	8	0.519	0.862	0.210	-0.030 (-0.043, -0.016)	< 0.001
pagerank-balance †	9	0.515	0.769	0.262	-0.035 (-0.058, -0.007)	0.012
repression-pagerank-balance †	7	0.486	0.928	0.718	-0.063 (-0.089, -0.034)	< 0.001
repression-dag-pagerank †	9	0.515	0.769	0.260	-0.035 (-0.066, -0.005)	0.020
repression-order-pagerank †	9	0.497	0.769	0.548	-0.053 (-0.084, -0.025)	< 0.001
tf-neighbour-share ‡	80	0.828	< 0.001	< 0.001	+0.279 (+0.231, +0.326)	< 0.001
tf-target-share ‡	75	0.812	< 0.001	< 0.001	+0.262 (+0.215, +0.308)	< 0.001
tf-sender-share ‡	78	0.831	< 0.001	< 0.001	+0.282 (+0.235, +0.328)	< 0.001
similar-targets ‡	70	0.751	< 0.001	< 0.001	+0.201 (+0.151, +0.251)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

gold

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	4,012	0.232 (10.47×)	9.53×	11.54×	0.97 (0.50)
resolve_reciprocals → drop_sinks → prune_triangles	3,289	0.096 (9.28×)	5.83×	11.81×	0.99 (0.61)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	1,112	0.085 (9.42×)	5.58×	11.74×	0.98 (0.65)
absolute	4,012	0.232 (10.47×)			
absolute → minimum_feedback_order	3,036	0.118 (9.19×)			
orient_reciprocals ratio=2.0 → absolute	2,162	0.019 (3.45×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	2,162	0.019 (3.45×)			
top_outgoing k=50 → absolute	4,012	0.231 (11.02×)			
top_outgoing k=50 → absolute → minimum_feedback_order	3,036	0.115 (9.93×)			
keep_sign sign=1 → absolute	4,012	0.112 (9.53×)			
keep_sign sign=1 → absolute → minimum_feedback_order	2,394	0.062 (8.88×)			
keep_sign sign=-1 → absolute	4,012	0.120 (11.54×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	2,784	0.092 (12.53×)			
keep_sign sign=-1	4,012	0.120 (11.54×)		11.54×	0.96 (0.96)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 60,389 edges (27,280 negative), largest strong component 1,643.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 60,389 edges (27,280 negative), largest strong component 1,643.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 29,436 edges (15,886 negative), largest strong component 1,258.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 29,436 edges (15,886 negative), largest strong component 1,258; DAG keeps 13,610 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 60,389 edges (0 negative), largest strong component 1,643; DAG keeps 37,435 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 60,389 edges (0 negative), largest strong component 1,643.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 60,389 edges (0 negative), largest strong component 1,643; DAG keeps 37,435 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 14,981 edges (0 negative), largest strong component 1; DAG keeps 14,981 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 56,255 edges (0 negative), largest strong component 1,643; DAG keeps 33,621 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 33,109 edges (0 negative), largest strong component 1,641; DAG keeps 20,159 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 27,280 edges (0 negative), largest strong component 1,628; DAG keeps 17,317 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 60,389 edges (27,280 negative), largest strong component 1,643.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 60,389 edges (27,280 negative), largest strong component 1,643.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 60,389 edges (27,280 negative), largest strong component 1,643.

reverse-pagerank: Highest reverse PageRank (raw graph)